# RENT-A-VACATION

## Technology & Tools Inventory

*Name Your Price. Book Your Paradise.*

**Ask RAVIO** — *Just Say Where. RAVIO Does the Rest.*

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**Date:** February 26, 2026
**Version:** 1.0
**Classification:** Internal — Business Operations
**Prepared by:** RAV Engineering Team

# 1. Subscription & Paid Services

These are the external SaaS products and services that RAV depends on. Most operate on free tiers during pre-launch; costs scale with usage post-launch.

| # | Service | Purpose | Free Tier | Paid Tier | Status |
|---|---|---|---|---|---|
| 1 | Claude Max (Anthropic) | AI coding assistant (Claude Code) | — | $100–200/mo | Active |
| 2 | Supabase | Database, Auth, Edge Functions, Storage | 500MB DB, 50K users, 500K invocations | $25/mo (Pro) | Active (DEV + PROD) |
| 3 | Vercel | Frontend hosting, CDN, preview deploys | 100GB bandwidth, 6K build min/mo | $20/mo (Pro) | Active |
| 4 | Stripe | Payments, Connect payouts, Tax | No platform fee | 2.9% + $0.30 per txn | Active |
| 5 | VAPI.ai | Voice assistant (Deepgram + ElevenLabs + OpenAI) | 10 min/mo | ~$0.05–0.15/min | Active |
| 6 | OpenRouter | Text chat LLM (Gemini 3 Flash) | Free tier models | ~$0.50/M tokens | Active |
| 7 | Resend | Transactional email | 3,000 emails/mo, 1 domain | $20/mo (5K+) | Active |
| 8 | Sentry | Error monitoring & performance | 5K errors/mo, 10K transactions | $26/mo (Team) | Active |
| 9 | Cloudflare | DNS, email routing (catch-all) | Free plan | — | Active (free) |
| 10 | Percy (BrowserStack) | Visual regression testing | 5K screenshots/mo | $399/mo (Team) | Active (CI) |
| 11 | Qase.io | Test case management & reporting | 500 test cases, 3 users | $36/mo (Startup) | Active (CI) |
| 12 | NewsAPI | Industry news feed (exec dashboard) | 100 req/day (dev only) | $449/mo (Business) | Active (dev key) |
| 13 | PostHog | Product analytics & user behavior | 1M events/mo | $1,600/yr (Scale Add-on) | Active (coupon thru |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | Jan 2027) |
| 14 | AirDNA | Vacation rental market intelligence (BYOK) | — | ~$250–500/mo (user-paid) | Planned — BYOK model |
| 15 | STR Global | Short-term rental benchmarks (BYOK) | — | Custom pricing (user-paid) | Planned — BYOK model |
| 16 | Canva | Marketing assets, brand design | Free tier | $13/mo (Pro) | Used for design |
| 17 | GitHub | Repo, Issues, Actions CI/CD | Free (public), 2K Actions min/mo | $4/user/mo (Team) | Active |

# 2. Integrated Providers

These are billed through a primary service listed above — you do not pay them separately.

| Provider | Billed Via | What It Does |
|---|---|---|
| Deepgram | VAPI | Speech-to-text (STT) |
| ElevenLabs | VAPI | Text-to-speech (TTS) |
| OpenAI GPT-4o-mini | VAPI | Voice assistant LLM |
| Google Gemini 3 Flash | OpenRouter | Text chat LLM |
| Google OAuth 2.0 | Supabase Auth | Social login (Sign in with Google) |
| PostgreSQL | Supabase | Relational database engine |
| Deno Deploy | Supabase | Edge function runtime |
| Let’s Encrypt | Vercel | SSL/TLS certificates |

# 3. Free External APIs

No account or subscription required for these services.

| Service | What It Does | Notes |
|---|---|---|
| FRED API (Federal Reserve) | Economic indicators for exec dashboard | Public API, no key required |
| Google Fonts | Roboto font family | CDN-hosted, free |
| Unsplash | Stock property & destination photos | Free for commercial use with attribution |

# 4. Not Yet Integrated in Code

These services have accounts, code scaffolding, or open GitHub issues but are not yet fully wired into the application.

| Service | Purpose | Integration Status | GitHub Issue |
|---|---|---|---|
| PostHog SDK | Frontend analytics integration | Account active (Scale plan), SDK not yet installed | #73 |
| Google Analytics (GA4) | Traffic analytics & marketing attribution | Not implemented | #74 |
| AirDNA | Live market data for exec dashboard | BYOK: edge function + settings UI exist, awaiting user API key | — |
| STR Global | Live rental benchmarks for exec dashboard | BYOK: edge function + settings UI exist, awaiting user API key | — |

> *BYOK (Bring Your Own Key): AirDNA and STR Global use a model where the admin enters their own API key via the Executive Dashboard → Integration Settings panel. RAV does not pay for these subscriptions — they are user-paid. The platform shows demo data until a key is provided.*

# 5. Open Source & Development Tools

Free, open-source tools used in the development stack. No subscription cost.

| Category | Tools |
|---|---|
| Frontend Framework | React 18, TypeScript 5.8, Vite 5.4 |
| Styling | Tailwind CSS 3.4, shadcn/ui (Radix UI primitives) |
| State & Data | TanStack React Query, React Hook Form, Zod |
| UI Components | Lucide icons, Recharts, Mermaid, date-fns, Embla Carousel |
| Testing | Vitest, Playwright, Testing Library, jsdom |
| Code Quality | ESLint 9, Husky, lint-staged |
| PWA | vite-plugin-pwa, Workbox (offline support, installable app) |

# 6. Monthly Cost Estimate (Pre-Launch)

Estimated costs while the platform is in pre-launch / Staff Only Mode with minimal traffic.

| Service | Estimated Cost | Notes |
|---|---|---|
| Claude Max | $100–200 | Primary development tool |
| Supabase (2 projects) | $0–50 | Free tier covers pre-launch |
| Vercel | $0–20 | Free tier likely sufficient initially |
| Stripe | $0 | Only charges per transaction |

| | | |
|---|---|---|
| VAPI | $0–10 | Minimal voice usage pre-launch |
| OpenRouter | $0–5 | Gemini Flash is very cheap |
| Resend | $0 | Free tier (3,000 emails/mo) |
| Sentry | $0 | Free tier (5K errors/mo) |
| Cloudflare | $0 | Free plan |
| Percy | $0 | Free tier for CI |
| Qase | $0 | Free tier |
| PostHog | $0 | Coupon until Jan 2027; then ~$133/mo ($1,600/yr) |
| AirDNA | $0 | BYOK — user-paid, not a RAV expense |
| STR Global | $0 | BYOK — user-paid, not a RAV expense |
| NewsAPI | $0 | Dev key (prod needs $449/mo or alternative) |
| GitHub | $0 | Free for current usage |
| Canva | $0–13 | Optional |
| TOTAL | $100–300/mo | Pre-launch; mostly Claude Max |

> *Post-launch costs scale with usage — mainly Stripe (per-transaction), Supabase (DB size + edge invocations), Vercel (bandwidth), and VAPI (voice minutes). PostHog coupon expires Jan 21, 2027 — then $1,600/yr. AirDNA and STR Global are BYOK (user-paid, not a RAV expense).*

# 7. Environment Configuration Summary

## Frontend Environment Variables (.env.local / Vercel)

| Variable | Service | Set In |
|---|---|---|
| VITE_SUPABASE_URL | Supabase | .env.local + Vercel |
| VITE_SUPABASE_ANON_KEY | Supabase | .env.local + Vercel |
| VITE_VAPI_PUBLIC_KEY | VAPI | .env.local + Vercel |
| VITE_VAPI_ASSISTANT_ID | VAPI | .env.local + Vercel |
| VITE_SENTRY_DSN | Sentry | .env.local + Vercel |
| VITE_FEATURE_VOICE_ENABLED | Feature flag | .env.local + Vercel |

## Backend Secrets (Supabase Edge Functions)

| Secret | Service | Used By |
|---|---|---|
| STRIPE_SECRET_KEY | Stripe | Checkout, payouts, refunds, webhooks |
| STRIPE_WEBHOOK_SECRET | Stripe | Webhook signature verification |
| RESEND_API_KEY | Resend | All transactional emails |
| OPENROUTER_API_KEY | OpenRouter | Text chat (RAVIO) |
| NEWSAPI_KEY | NewsAPI | Industry news feed (optional) |
| SUPABASE_SERVICE_ROLE_KEY | Supabase | Admin operations in edge functions |
| IS_DEV_ENVIRONMENT | Internal | Guards seed data in production |

## CI/CD Secrets (GitHub Actions)

| Secret | Service | Used By |
|---|---|---|
| PERCY_TOKEN | Percy | Visual regression tests |
| QASE_API_TOKEN | Qase | Test reporting |
| SUPABASE_URL | Supabase | CI test environment |
| SUPABASE_ANON_KEY | Supabase | CI test environment |
| RESEND_GITHUB_NOTIFICATIONS_KEY | Resend | Issue email notifications |

# 8. Architecture Overview

| Layer | Technology | Details |
|---|---|---|
| Frontend | React + TypeScript + Vite | SPA deployed on Vercel CDN |
| UI Library | Tailwind CSS + shadcn/ui | Utility-first CSS + Radix primitives |
| Backend | Supabase | PostgreSQL + PostgREST API + Auth + Edge Functions |
| Edge Functions | Deno (TypeScript) | 24 serverless functions for business logic |
| Payments | Stripe | Checkout, Connect (owner payouts), webhooks, tax |
| Voice AI | VAPI → Deepgram + ElevenLabs + OpenAI | Browser-based voice search |
| Text AI | OpenRouter → Gemini 3 Flash | Conversational assistant (RAVIO) |
| Email | Resend | Transactional emails (7 templates) |
| Monitoring | Sentry | Error tracking + performance |
| Analytics | PostHog | Product analytics + user behavior (Scale plan) |
| Market Data | AirDNA + STR Global | Rental market intelligence (planned) |
| DNS / CDN | Cloudflare + Vercel | DNS routing, email catch-all, edge CDN |
| CI/CD | GitHub Actions | Lint, test, visual regression, deploy |
| Testing | Vitest + Playwright + Percy + Qase | Unit, E2E, visual, reporting |